

# 4.3.30

EE25BTECH11041 - Naman Kumar

Question:

Find the equation of the line which passes through the point  $(-4, 3)$  and the portion of the line intercepted between the axes is divided internally in ratio 5:3 by this point.

**Solution:**

Let the intercept points be

$$\mathbf{P} = a\mathbf{e}_1, \mathbf{Q} = b\mathbf{e}_2 \left( \mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, a \text{ and } b \text{ are constants} \right) \quad (1)$$

and

$$\mathbf{R} = \begin{pmatrix} -4 \\ 3 \end{pmatrix} \quad (2)$$

be the given point.

Using

$$\mathbf{R} = \frac{k\mathbf{Q} + \mathbf{P}}{k + 1} \quad (3)$$

$$\mathbf{R} = \frac{5 \times b\mathbf{e}_2 + 3 \times a\mathbf{e}_1}{8} \quad (4)$$

$$\begin{pmatrix} -32 \\ 24 \end{pmatrix} = \begin{pmatrix} 3a \\ 5b \end{pmatrix} \quad (5)$$

General equation of line

$$\mathbf{x} = \mathbf{h} + c\mathbf{m} \quad (6)$$

Where

|          |                        |
|----------|------------------------|
| <b>x</b> | general vector on line |
| <b>h</b> | known vector of line   |
| <b>m</b> | slope vector of line   |
| <b>c</b> | scalar parameter       |

TABLE I

Slope is

$$\mathbf{m} = \mathbf{Q} - \mathbf{P} \quad (7)$$

$$(8)$$

let  $\mathbf{h} = \mathbf{P}$

So, Equation of line is

$$\mathbf{x} = \mathbf{h} + c\mathbf{m} \quad (9)$$

$$\mathbf{x} = \mathbf{P} + c(\mathbf{Q} - \mathbf{P}) \quad (10)$$

Putting values of  $\mathbf{Q}, \mathbf{P}$

$$\mathbf{x} = a\mathbf{e}_1 + c \begin{pmatrix} -a \\ b \end{pmatrix} \quad (11)$$

$$\mathbf{x} = \begin{pmatrix} \frac{-32}{3} \\ 0 \end{pmatrix} + c \begin{pmatrix} \frac{32}{3} \\ \frac{24}{5} \end{pmatrix} \quad (12)$$

Therefore Final equation is

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{-32}{3} \\ 0 \end{pmatrix} + k \begin{pmatrix} \frac{-32}{3} \\ \frac{24}{5} \end{pmatrix} \quad (13)$$

